

# Machine Learning Approach to Forecasting Nonprofit Donor Retention

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**Abstract:** Donor lapse remains a persistent financial challenge for nonprofits, where the cost of reacquiring a lapsed donor substantially exceeds the cost of proactive retention. This project develops a machine learning pipeline to predict year-conditional donor lapse from historical giving behavior using data from Curebound, a San Diego-based nonprofit raising funds for cancer research. We frame the problem as a binary classification task on an annual donor-by-year panel constructed from over 221,000 cleaned Salesforce CRM transactions spanning 2013-2026. Leakage-safe features derived exclusively from prior-year giving history are used to train three progressively complex classifiers (logistic regression, random forest, and XGBoost) under a strict temporal train/validation/test split (2014-2023/2024/2025). Probability calibration via Platt scaling is applied to XGBoost to ensure that output risk scores are meaningful for operational decision-making. XGBoost achieves the strongest overall performance (AUPRC  $\approx 0.75$ ), with its advantage most pronounced in the top-K outreach regime, where targeting the top 1,000 highest-risk donors yields approximately 79% precision while capturing roughly 26% of all actual lapses, representing a 1.26x lift over random outreach. These results demonstrate that a well-designed ML pipeline can improve donor retention targeting under realistic resource constraints.

**Keywords:** donor lapse prediction, gradient boosting, temporal validation, precision-recall

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## 1. Introduction

Nonprofit organizations depend on consistent donor support to sustain their programs, fund staffing, and plan long-term initiatives. Yet donor lapse, the failure of a previously active donor to contribute in a given year, is widespread across the sector. Research consistently shows that once a donor disengages, the cost of reacquisition significantly exceeds the cost of retaining them in the first place. This dynamic transforms proactive donor retention from a nice-to-have into a financial necessity, particularly for organizations operating with limited outreach capacity and lean staff.

The conventional response to lapse risk has been static CRM rules: flag a donor if they have not given in 18 months, then trigger a reactivation email. These rules are easy to implement but fail to capture the complexity and heterogeneity of real donor behavior. A major donor who skips one year after a decade of giving looks very different from a one-time small contributor who never returned, yet both might trigger the same rule-based alert. Untargeted mass outreach

wastes budget and staff attention, and may even erode donor relationships if the communication feels tone-deaf.

Machine learning offers a principled alternative: rather than applying a single threshold to all donors, it can learn from historical patterns to assign each donor a calibrated risk score, enabling organizations to rank and prioritize outreach in a way that is both targeted and resource-aware. This project asks a direct operational question: can we predict year-conditional donor lapse from historical giving behavior in a way that is actionable under real nonprofit constraints?

The input to our pipeline is a donor-by-year panel derived from opportunity-level donation records, with features constructed exclusively from prior-year giving history (e.g. lifetime gift count through the previous year, last year's gift count and total amount) The output is a calibrated probability score indicating each eligible donor's likelihood of lapsing in the prediction year. This score is used to rank donors by risk, enabling the nonprofit to contact the

highest-risk individuals first given whatever outreach capacity is available.

## 2. Related Work

Prior work on donor retention and churn prediction can be grouped into three broad categories: heuristic scoring models, traditional supervised classification, and more recent machine learning approaches applied to behavioral data.

### **Heuristic & Rule-Based Approaches:**

The dominant industry framework for donor segmentation has long been RFM analysis: scoring donors on recency, frequency, and monetary value. RFM is intuitive and easy to explain to nonprofit staff, and it captures some signal about lapse risk. However, its static thresholds cannot adapt to individual donor trajectories, and it ignores interactions between features. Rule-based CRM systems (e.g. send a reactivation email after 18 months of inactivity) suffer from similar rigidity: they produce binary flags rather than ranked risk scores, and they fail to differentiate among donors with widely varying giving histories.

### **Traditional Statistical Models:**

Survival analysis and logistic regression have been applied to donor churn, offering interpretability and a probabilistic framing. These approaches represent a meaningful step beyond heuristics, as they can incorporate multiple predictors simultaneously and provide calibrated probabilities. However, logistic regression assumes linear decision boundaries, which may not capture the nonlinear relationship between donor engagement history and lapse probability that our EDA confirms exists.

### **Machine Learning on Behavioral Data:**

Althoff and Leskovec [1] studied donor retention on DonorsChoose.org using logistic regression, random forest, and SVM across 470,000 donors, finding that roughly 74% of donors never returned after their first gift and that timely post-donation communication meaningfully improved retention. Schetgen et al. [2]

demonstrated that Facebook social media data (specifically liked pages and page categories) can predict first-time donation behavior with meaningful accuracy ( $AUC = 0.72$ ), offering nonprofits an acquisition modeling approach that does not require a pre-existing donor relationship. Hesse [3] applied random forest to a peer-to-peer fundraising context, achieving 79% prediction accuracy in identifying which peer donors would convert to organizational donors, and found via logistic regression that post-donation communication was associated with a 4x increase in the odds of conversion.

### **Gaps Addressed by This Work:**

Despite this progress, several consistent methodological gaps persist. First, most studies define lapse as a static, permanent label (a donor is "lapsed" once they stop giving) rather than treating it as a year-conditional behavior that can cycle. Second, many studies fail to enforce temporal train/test splits, creating feature leakage that inflates reported performance. Third, most prior work optimizes for ROC AUC rather than AUPRC, which is more meaningful under class imbalance and more closely aligned with the decision of who to contact. Finally, most studies focus on first or second-time donors rather than the sustained multi-year relationships that are most valuable to nonprofits. Our project directly addresses each of these gaps through a year-conditional lapse definition, a strict temporal split, AUPRC-centric evaluation, and a focus on donors with established giving histories.

## 3. Dataset & Features

### **Data Source & Construction:**

Our dataset originates from Curebound, a San Diego-based nonprofit dedicated to accelerating cancer research through philanthropic fundraising. The raw data consists of three internal reports exported from Salesforce CRM and joined via SQL on a unique donor identifier. After merging and deduplication, the cleaned dataset contains over 221,000 opportunity-level transaction records spanning 2013 through 2026, covering 23 columns including transaction

amount, transaction date, campaign engagement indicators, and donor account metadata.

All personally identifiable information (PII) was removed prior to any analysis, as the donor account ID (already present in the CRM) was sufficient as a unique key. Missing data was negligible across all core modeling fields; no imputation was required. The cleaning pipeline enforces type consistency on `close_date` and amount fields, removes duplicate opportunity IDs, and flags non-positive amounts for review. Figure 1 shows the missingness summary and donation volume by year, illustrating both the data quality and the long-term giving trends across the dataset.

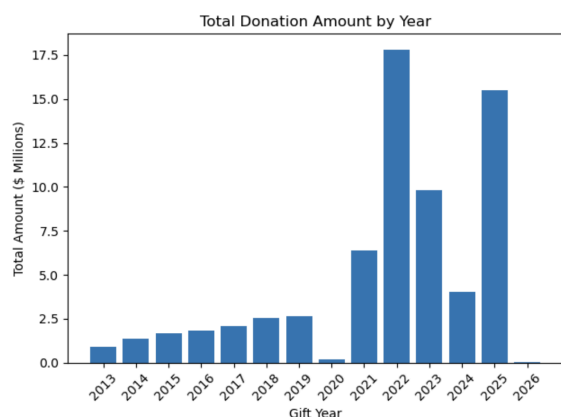


Figure 1: Total donation amount by year (2013-2026) showing sustained growth in giving volume with a notable decline in 2020 consistent with COVID-19.

### Panel Construction & Lapse Labeling:

Rather than modeling at the transaction level, we construct a donor-by-year panel to enable year-conditional lapse prediction. For each donor and each calendar year, we aggregate transaction records to produce annual summaries. Critically, we explicitly materialize "no-donation" rows for years in which an otherwise-active donor made no contributions; this is essential for detecting lapse as an absence of giving rather than merely a lack of records.

A donor is defined as lapsed in year Y if they (1) gave at least once in year Y-1, (2) have at least two lifetime gifts through year Y-1, and (3) do not give in year Y. This year-conditional definition has two important properties. First, it

is dynamic: a donor who lapsed in 2019 but returned in 2020 is labeled as retained in 2020, accurately reflecting the cyclical nature of donor behavior. Second, the eligibility criteria (at least two lifetime gifts, active the prior year) focus the model on the population of donors for whom a meaningful prediction can be made, avoiding the noise introduced by one-time donors or those who never had a giving relationship with the non-profit organization.

The overall lapse rate across eligible donor-years ranges from roughly 45% to 97% annually, with an elevated rate in 2020 consistent with COVID-19 disruptions. The current 2025 test-year lapse rate among eligible accounts is approximately 63%.

### Leakage-Safe Feature Engineering:

All features used in modeling are constructed using only information available before the prediction year, using temporal shifts of the donor-year panel. The core feature set includes: (1) `donation_year`: the calendar year of prediction, included to capture temporal trends; (2) `lifetime_gifts_through_last_year`: the total number of gifts through the end of year Y-1; (3) `gift_count_last_year`: the number of gifts made in Y-1; and (4) `total_amount_last_year`: total dollars donated in Y-1.

Several EDA findings directly shaped our feature construction and modeling choices. First, plotting the lapse rate against lifetime gift count revealed a sharply nonlinear decline, as shown in Figure 2: donors with only two lifetime gifts lapse at rates above 70%, while those with 100+ lifetime gifts lapse below 10%. Because a linear model cannot capture this curvature, this finding motivated our decision to include nonlinear ensemble methods (random forest and XGBoost) alongside logistic regression rather than treating the baseline as sufficient. Second, EDA showed that donation volume dropped dramatically in 2020 (consistent with COVID-19 disruptions) with a lapse rate near 97% for that year among eligible accounts. This confirmed that year-specific effects exist in the data and reinforced the decision to use a strict temporal train/val/test split rather than random cross-validation, which would risk leaking

post-2020 recovery patterns into training. Third, the distribution of gift counts per donor is heavily right-skewed, with a small number of donors giving many times per year. Because tree-based models handle skewed distributions without requiring log-transformation while linear models benefit from it, this EDA finding informed our decision to apply feature scaling only to logistic regression and not to random forest or XGBoost. Finally, the negligible missingness across all core features (under 0.01% for gift count, last year amount, and first gift date) confirmed that no imputation strategy was necessary and that the feature set could be used as-is without introducing imputation-related variance into the pipeline.

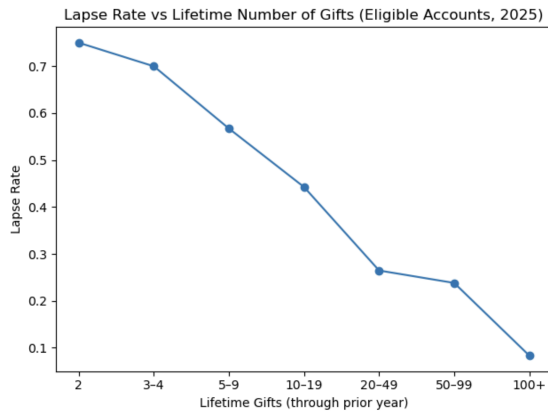


Figure 2: Lapse rate by lifetime gift count bucket for eligible accounts (2025), showing a nonlinear decline in lapse probability with cumulative giving tenure.

## 4. Methods

### Modeling Strategy:

We frame donor lapse prediction as a binary classification task. For each eligible donor-year record in the panel, the model estimates the probability that the donor will lapse in the following year. This probabilistic output, rather than a hard binary label, is central to the operational goal: nonprofits need to rank donors by risk so they can allocate limited outreach capacity to the highest-priority individuals.

To prevent temporal leakage and simulate realistic deployment conditions, we enforce a strict temporal split. The training set covers

donor-year records from 2014 through 2023; the validation set covers 2024; and the true holdout test set covers 2025. Models are trained on the training set, hyperparameters are selected based on validation performance, and final evaluation is reported exclusively on the 2025 holdout. Under no circumstances does any future info appear in the features of an earlier record.

Class imbalance (lapsed donors outnumber retained donors in most years) is addressed differently across models: logistic regression uses class-weighted loss, random forest uses balanced subsampling, and XGBoost uses its native `scale_pos_weight` parameter. Probability calibration via Platt scaling (a sigmoid fit on validation set outputs) is applied to XGBoost before final evaluation to ensure that predicted scores reflect empirically observed lapse rates. All models were implemented in Python using scikit-learn [8].

### Logistic Regression (Baseline):

Logistic regression serves as our interpretable baseline. The model estimates the log-odds of lapse as a linear combination of the feature vector  $x$ :

$$\log[p/(1-p)] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Features are standardized via z-score normalization prior to fitting. The regularization strength  $C$  is tuned over a logarithmic grid using validation AUPRC. Class imbalance is addressed by weighting the loss function inversely proportional to class frequencies. Logistic regression's transparency makes it useful as a performance floor and as a reference point for understanding how much of the signal can be captured linearly.

### Random Forest:

Random forest [4] builds an ensemble of decision trees, each trained on a bootstrapped subsample of the data with a random subset of features considered at each split. The ensemble prediction is the average of individual tree probabilities. This approach captures nonlinear relationships and feature interactions that logistic regression cannot represent. To handle

class imbalance, we use balanced subsampling, which resamples each bootstrap so that the lapsed and retained classes are equally represented. The number of estimators and maximum depth are tuned on the validation set. Random forest serves as an intermediate-complexity model between logistic regression and XGBoost.

### XGBoost (Primary Model):

XGBoost [5] implements gradient boosted decision trees, iteratively fitting shallow trees to the residuals of the current ensemble. Formally, at iteration  $t$ , the model adds a tree  $f_t$  that minimizes the regularized objective:

$$L(t) = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t)$$

where  $l$  is the binary cross-entropy loss and  $\Omega$  penalizes tree complexity (leaf count and L2 regularization on leaf weights). XGBoost natively handles class imbalance via the `scale_pos_weight` parameter, set to the ratio of negative to positive instances. Early stopping on validation AUPRC prevents overfitting during training. After identifying the optimal number of rounds, the model is refitted on the combined train + validation data before final evaluation on the 2025 holdout. XGBoost is our primary model because gradient boosting consistently outperforms other methods on structured tabular data with moderate feature dimensionality, as confirmed by prior donor churn literature and our own validation results.

### Probability Calibration:

Raw ensemble model scores are often poorly calibrated, a predicted probability of 0.8 does not reliably correspond to an 80% empirical lapse rate. Because the nonprofit's use case requires trustworthy risk scores (not just rank ordering), we apply Platt scaling [6] to XGBoost. Platt scaling fits a sigmoid function on the validation set outputs, mapping raw model scores to calibrated probabilities. The resulting calibration curve for XGBoost, as shown in Figure 3, confirms substantial alignment between predicted and observed rates across most of the probability range.

## 5. Experiments / Results / Discussion

### Evaluation Metrics:

We use AUPRC as the primary evaluation metric. AUPRC summarizes model performance across all possible classification thresholds by computing the area under the precision-recall curve [7], where precision =  $TP/(TP + FP)$  and recall =  $TP/(TP + FN)$ . AUPRC is preferable to ROC AUC under class imbalance because ROC AUC is insensitive to the proportion of positive instances. We additionally report precision, recall, and F1-score at fixed thresholds, and evaluate operational performance using top-K metrics: given a budget to contact  $K$  donors, we rank all eligible donors by predicted risk and compute precision@ $K$ , recall@ $K$ , and lift@ $K$  (ratio of precision@ $K$  to the base lapse rate) as  $K$  varies from 0 to 5,000.

### Model Performance:

| Model          | AUPRC | p@1K  | r@1K  | lift@1K |
|----------------|-------|-------|-------|---------|
| Logistic Reg.  | 0.747 | —     | —     | —       |
| Random Forest  | 0.745 | —     | —     | —       |
| XGBoost (Cal.) | 0.754 | 0.793 | 0.255 | 1.263   |

Table 1: Test set performance (2025 holdout).

XGBoost achieves the highest AUPRC of 0.754 on the 2025 test holdout, though all three models perform within a narrow band. Logistic regression achieves 0.747 and random forest 0.745, demonstrating that even a linear model extracts meaningful signals from this compact feature set. The more meaningful separation emerges in the top-K regime, where XGBoost's higher precision at small  $K$  translates directly into fewer wasted outreach contacts. As shown in Figure 3, the calibration curve for XGBoost post-Platt scaling shows good agreement between predicted and observed lapse rates with mild overestimation in the high-probability range, a conservative bias that is arguably preferable in a retention context.

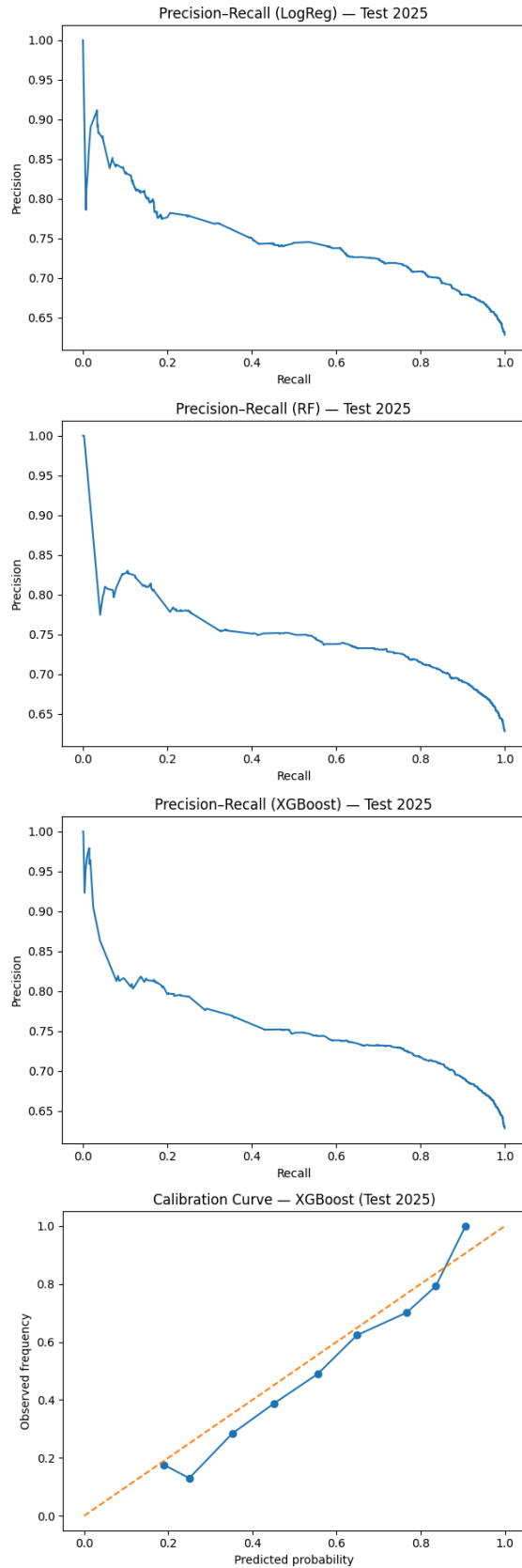
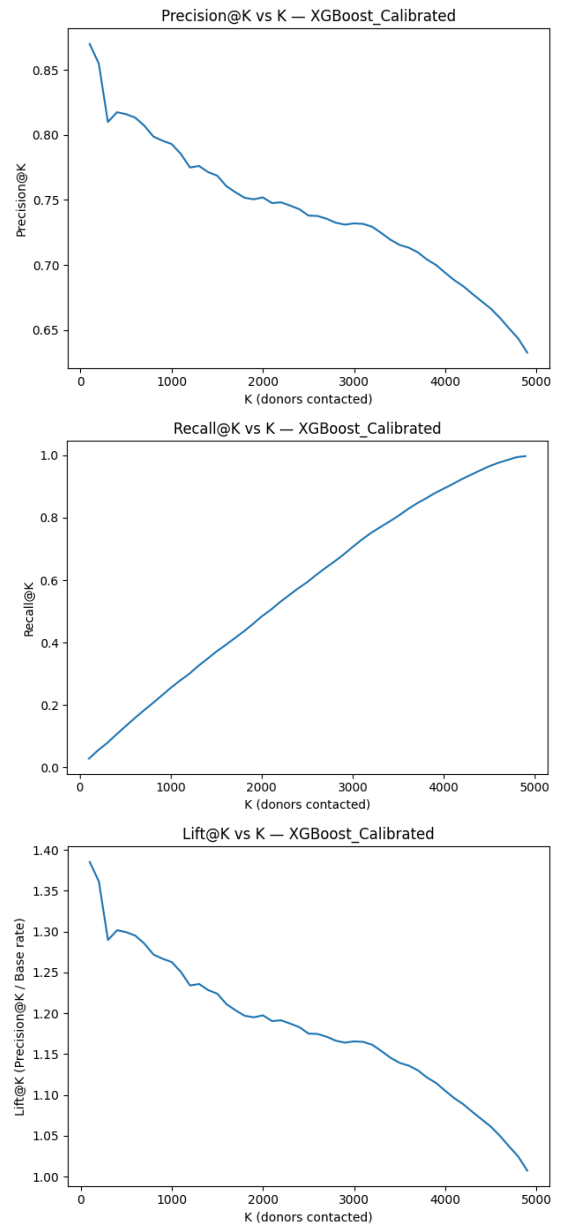


Figure 3: Precision-recall curves for logistic regression, random forest, and XGBoost (top to bottom), with XGBoost calibration curve (bottom).

### Operational Evaluation Top-K Outreach:

Targeting the top 1,000 highest-risk donors captures approximately 26% of all actual lapses at 79% precision; roughly 4 out of 5 flagged donors are genuinely at risk. As shown in Figure 4, lift@K peaks at small K (approximately 1.35 at K=100-200) and converges toward 1.0 as K approaches the full donor pool, meaning the model delivers the most value precisely when outreach capacity is most limited.



*Figure 4: precision@K, recall@K, and lift@K for XGBoost (calibrated), 2025 holdout; lift is highest at small K where outreach capacity is most constrained.*

### **Cross-Model Risk Ranking Consistency:**

To verify that the predictive signal is robust rather than model-specific, we examined agreement across all three models on the 2025 holdout. All three achieve AUPRC within a narrow band (logistic regression = 0.747, random forest = 0.745, XGBoost = 0.754). This convergence across a linear model, a bagged ensemble, and a boosted ensemble suggests that the core signal, driven primarily by lifetime gift count and prior-year giving activity, is model-agnostic and stable. XGBoost's advantage is most meaningful at small K, where its higher precision translates directly into fewer wasted outreach contacts, rather than in overall ranking agreement.

### **Ensemble Methods v. Heuristics:**

The claim that ensemble methods outperform heuristics is supported both by prior literature and by our own comparative results. Althoff and Leskovec [1] showed that even simple ML models could predict donor return with meaningful accuracy (ROC AUC = 0.74) using first-donation features alone; Schetgen et al. [2] demonstrated that ensemble methods such as XGBoost and random forest were competitive with logistic regression when applied to social media data for donor acquisition; and Hesse [3] found that random forest slightly outperformed logistic regression in predicting peer donor conversion (79% vs. 77% accuracy). In our experiment, all three models achieve similar AUPRC on the 2025 holdout (logistic regression = 0.747, random forest = 0.745, XGBoost = 0.754) though XGBoost's advantage is more pronounced in the top-K regime, where its higher precision at small K translates directly into fewer wasted outreach contacts.

### **Error Analysis:**

False negatives concentrate disproportionately among long-tenured, high-lifetime gift donors (a pattern that makes intuitive sense), as a decade-long consistent donor looks behaviorally

indistinguishable from one about to lapse, and a single missed year may reflect situational factors rather than genuine disengagement. This represents the core limit of transactional data alone; incorporating engagement signals such as email open rates and event attendance is a clear direction for future work.

## **6. Conclusion**

This project demonstrates that a carefully designed machine learning pipeline can substantially improve nonprofit donor retention targeting compared to conventional rule-based approaches. Working with over 221,000 real donation records from Curebound spanning more than a decade, we constructed a leakage-safe donor-by-year panel, defined a year-conditional lapse label that captures the cyclical nature of donor behavior, and trained three classifiers under a strict temporal evaluation framework. XGBoost emerged as the strongest performer, achieving AUPRC = 0.754 on the 2025 holdout and enabling the nonprofit to contact the 1,000 highest-risk donors at approximately 79% precision, a 1.26x lift over random outreach. Probability calibration via Platt scaling ensured these scores are trustworthy for real operational decisions.

Looking back at project inception goals, we largely delivered: a reproducible leakage-safe pipeline, year-conditional lapse labeling, temporal train/test splitting, AUPRC-centered evaluation, and an operationally actionable top-K analysis. The primary gap is the limited feature scope, working exclusively with transactional data rather than broader donor engagement signals. Future iterations should prioritize incorporating these behavioral signals, adding recency-gap and donation-volatility features, and exploring uplift modeling to shift the question from who is at risk to who will actually respond to outreach, a more actionable framing for real nonprofit decision-making. At a broader level, this work demonstrates how machine learning can enable nonprofits to transition from reactive fundraising to proactive donor stewardship, strengthening donor relationships and supporting the long-term sustainability of mission-driven organizations.

## Contributions

Julian Estrada led data acquisition and preprocessing, including SQL joins across three Salesforce CRM exports, canonical cleaning, and donor-by-year panel construction. Julia Klayman led the literature review and project framing, synthesizing prior work on RFM heuristics, supervised churn modeling, and identifying methodological gaps addressed by this work. Tristan Roman led model development and the code pipeline, implementing logistic regression, random forest, and XGBoost with probability calibration in a reproducible Jupyter notebook. David Lightfoot led results analysis and operational evaluation, designing the top-K outreach framework and interpreting model outputs in the context of nonprofit resource constraints. All four members contributed to the writing and editing of this report.

## Reflections

If we had the opportunity to redo this project, the most consequential change would be to begin the data integration conversation with Curebound earlier, with the explicit goal of incorporating donor engagement data (email open rates, event attendance, campaign response history) in addition to transactional records. Our EDA confirmed that long-tenured donors are the hardest subpopulation to predict using transaction history alone, and behavioral engagement signals would most directly address this gap. We would also explore uplift modeling from the outset, as the shift from predicting who will lapse to who will respond to outreach more directly answers the question a nonprofit actually needs to act on. On the technical side, we would invest earlier in a more systematic hyperparameter search and feature importance analysis, both to improve model performance and to better explain to a non-technical nonprofit audience which signals are driving the predictions. Finally, we would create clearer illustrative donor archetypes during the video presentation to make the lapse labeling logic more immediately intuitive, as noted by peers.

## References

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## Reply to Peer Reviews

### **Response to Group 9:**

(W1) Limitations & Lapse Cycle: Group 9 noted that limitations were concentrated at the end of the video presentation for the second progress report. In this report, we have woven limitation discussions into the relevant sections. We have also clarified the lapse cycle directly: a donor who lapsed in year Y can be labeled retained in year Y+1 if they give again, and our panel explicitly captures this by assigning a new label each year independently. This is one of the core distinctions between our year-conditional definition and the static lapse labels used in prior literature.

(W2) Dataset Granularity: The raw data is at the opportunity (transaction) level, each row represents a single donation event. We aggregate these transactions to the donor-by-year level as the first step of our panel construction, described in detail in the Dataset & Features section above.

(W3) Feature Importance: Lifetime gifts through the prior year and last year's gift count are the strongest individual predictors, consistent with the EDA finding that lapse probability declines sharply with cumulative giving tenure. A SHAP value analysis or permutation importance visualization would be a valuable addition to a future iteration of this work.

### **Response to Group 12:**

(W1) Lapse Labeling Clarity: To make the logic concrete, a donor who gives in 2013 and 2014 but not 2015 is labeled lapsed in 2015. If they give again in 2017 (having given in 2016), they are labeled retained in 2017. Two donations in one year count as two lifetime gifts for eligibility purposes; what matters is cumulative gift count, not years active; gift count within a year is captured separately as a feature. We have added a clear explanation of these mechanics in the Dataset & Features section.

(W2) Model Interpretability for Nonprofits: The calibrated probability scores from XGBoost can be explained to a nonprofit in operational terms: this donor has a 78% estimated chance of not giving this year. In a future iteration of this project, we would add a SHAP summary plot to illustrate which features drive individual predictions, making the model's decision-making transparent to non-data scientists/staff.

(W3) Dataset Context & Feature Explanation: The features warranted more introduction during the video presentation. The opportunity\_id is simply a unique transaction identifier used for deduplication before aggregation; it does not appear in the final model. In this report, we have described the feature set more thoroughly in the Dataset & Features section, including the meaning of each leakage-safe feature as well.